

Project 4: Implementing a Database

Due date: 25 March 2026 before 9 p.m.

In this project, students are required to implement a database with respect to the normalized logical data model represented in the format of a relational model.

Things to do

Use SQL to implement your database with respect to the normalized relational schema and **constraints**. The implementation includes (1) **Database Creation SQL script** which will be used to destroy the database if it already exists and then create a new database and constraints and (2) **Database Loading SQL script** to load data into the database with appropriate data type checking. The number of records to be loaded into each table should be appropriate. For example, the number of records in Branch table should be in the scale of tens to hundreds; the number of records in Staff and that in Client tables should be in the scale of hundreds to thousands; and the number records in Viewing table (i.e., transaction table) should be in the scale of thousands to millions. You are also required to write another **SQL script to report the number of records loaded into the database**. All scripts should be runnable on MS-SQL Server.

Concerned Issues

1. Plagiarism is NOT allowed for any cases.
2. Creativity in design and presentation is preferable.

Deliverables

A document in portable document format (PDF). The content in the document includes:

1. A document in portable document format (PDF). The content in the document includes:
 - a. The relational database schema from phase 3 and the corresponding ER diagram.
 - b. A discussion on the implementation of your database including difficulty, challenges, solutions, limitation.
2. The Database Creation SQL script.
3. The Database Loading SQL script for data loading and all data files (if any).
4. The SQL script for reporting the number of records in each table.

Submissions

1. If there are more than one file to be submitted, zip them. Then, name the file as <ProjectID>_<GID>_<last-three-digit-of-StudentID for every team member>_report.pdf, for example, "**P4_G01_999_988_977.zip**".
2. Submit the file on MyCourses: <https://mycourses.ict.mahidol.ac.th/>